Chapter 1. Primeiros passos

- Carregar bibliotecas e configurações comuns
- Carregar bibliotecas adicionais

A previsão do tempo fascina as pessoas há milhares de anos, sendo por vezes considerada um sinal de inspiração divina e por vezes vista como uma atividade criminosa. O profeta judeu Isaías escreveu por volta de 700 a.C.

> *Diga-nos o que o futuro nos reserva, para que saibamos que vocês são deuses.*
> (Isaías 41:23)

Cem anos depois, na antiga Babilônia, os adivinhos previam o futuro com base na aparência do fígado de uma ovelha. Quase na mesma época, as pessoas que buscavam previsões viajavam até Delfos, na Grécia, para consultar o Oráculo, que fornecia suas previsões sob o efeito de vapores de etileno. Os adivinhos enfrentaram maiores dificuldades sob o imperador Constâncio II, que emitiu um decreto em 357 d.C. proibindo qualquer pessoa de "consultar um adivinho, um matemático ou um vidente... Que a curiosidade de predizer o futuro seja silenciada para sempre".[1] Uma proibição semelhante à prática de previsões ocorreu na Inglaterra em 1824[2], quando "toda pessoa que fingisse ou afirmasse ler a sorte" foi "considerada um patife e vagabundo". A punição era de até três meses de prisão com trabalhos forçados!

A variabilidade na sorte dos analistas meteorológicos decorre do fato de que boas previsões podem parecer quase mágicas, enquanto previsões ruins podem ser perigosas. Considere as seguintes previsões famosas sobre computação.

- *Acho que existe um mercado mundial para talvez cinco computadores.* (Presidente da IBM, 1943)
- *Os computadores do futuro poderão pesar no máximo 1,5 toneladas.* (Popular Mechanics, 1949)
- *Não há razão para alguém querer um computador em casa.* (Presidente, DEC, 1977)

O último desses computadores foi fabricado apenas três anos antes de a IBM produzir o primeiro computador pessoal. Não é de surpreender que você não possa mais comprar um computador DEC. Fazer previsões é obviamente uma atividade difícil, e as empresas que a fazem bem têm uma grande vantagem sobre aquelas cujas previsões falham.

Neste livro, exploraremos os métodos mais confiáveis para gerar previsões usando Python. A ênfase será em métodos replicáveis, testáveis e que já tenham demonstrado eficácia.

1.1 O que pode ser previsto?

A previsão é necessária em muitas situações: decidir se deve construir outra usina de geração de energia nos próximos cinco anos exige previsões da demanda futura; programar a equipe de um call center na próxima semana exige previsões do volume de chamadas; abastecer um estoque exige previsões das necessidades de estoque. As previsões podem ser necessárias com vários anos de antecedência (no caso de investimentos de capital) ou apenas alguns minutos antes (para roteamento de telecomunicações). Independentemente das circunstâncias ou dos horizontes temporais envolvidos, a previsão é uma importante ferramenta para um planejamento eficaz e eficiente.

Algumas coisas são mais fáceis de prever do que outras. O horário do nascer do sol amanhã de manhã pode ser previsto com precisão. Por outro lado, os números da loteria de amanhã não podem ser previstos com qualquer exatidão. A previsibilidade de um evento ou de uma quantidade depende de diversos fatores, incluindo:

1. O quão bem compreendemos os fatores que contribuem para isso;
2. Quantos dados estão disponíveis?
3. Quão semelhante o futuro é ao passado;
4. se as previsões podem afetar aquilo que estamos tentando prever.

Por exemplo, as previsões de curto prazo da demanda residencial de eletricidade podem ser bastante precisas, pois geralmente todas as quatro condições são satisfeitas.

1. Temos uma boa noção dos fatores que contribuem para isso: a demanda por eletricidade é impulsionada principalmente pelas temperaturas, com efeitos menores para variações no calendário, como feriados, e condições econômicas.
2. Geralmente, estão disponíveis dados sobre a demanda de eletricidade de vários anos, e dados sobre as condições climáticas de muitas décadas.
3. Para previsões de curto prazo (até algumas semanas), é seguro assumir que o comportamento da demanda será semelhante ao observado no passado.
4. Para a maioria dos consumidores residenciais, o preço da eletricidade não depende da demanda, e, portanto, as previsões de demanda têm pouco ou nenhum efeito sobre o comportamento do consumidor.

Desde que tenhamos as habilidades para desenvolver um bom modelo que relacione a demanda de eletricidade com as principais variáveis determinantes, as previsões podem ser notavelmente precisas.

On the other hand, when forecasting currency exchange rates, only one of the conditions is satisfied: there is plenty of available data. However, we have a limited understanding of the factors that affect exchange rates, the future may well be different to the past if there is a financial or political crisis in one of the countries, and forecasts of the exchange rate have a direct effect on the rates themselves. If there are well-publicised forecasts that the exchange rate will increase, then people will immediately adjust the price they are willing to pay and so the forecasts are self-fulfilling. In a sense, the exchange rates become their own forecasts. This is an example of the "efficient market hypothesis". Consequently, forecasting whether the exchange rate will rise or fall tomorrow is about as predictable as forecasting whether a tossed coin will come down as a head or a tail. In both situations, you will be correct about 50% of the time, whatever you forecast. In situations like this, forecasters need to be aware of their own limitations, and not claim more than is possible.

Often in forecasting, a key step is knowing when something can be forecast accurately, and when forecasts will be no better than tossing a coin. Good forecasts capture the genuine patterns and relationships which exist in the historical data, but do not replicate past events that will not occur again. In this book, we will learn how to tell the difference between a random fluctuation in the past data that should be ignored, and a genuine pattern that should be modelled and extrapolated.

Many people wrongly assume that forecasts are not possible in a changing environment. Every environment is changing, and a good forecasting model captures the way in which things are changing. Forecasts rarely assume that the environment is unchanging. What is normally assumed is that *the way in which the environment is changing* will continue into the future. That is, a highly volatile environment will continue to be highly volatile; a business with fluctuating sales will continue to have fluctuating sales; and an economy that has gone through booms and busts will continue to go through booms and busts. A forecasting model is intended to capture the way things move, not just where things are. As Abraham Lincoln said, "If we could first know where we are and whither we are tending, we could better judge what to do and how to do it".

Forecasting situations vary widely in their time horizons, factors determining actual outcomes, types of data patterns, and many other aspects. Forecasting methods can be simple, such as using the most recent observation as a forecast (which is called the **naïve method**), or highly complex, such as neural nets and econometric systems of simultaneous equations. Sometimes, there will be no data available at all. For example, we may wish to forecast the sales of a new product in its first year, but there are obviously no data to work with. In situations like this, we use judgmental forecasting, discussed in Chapter 6. The choice of method depends on what data are available and the predictability of the quantity to be forecast.

1.2 Forecasting, goals and planning

Forecasting is a common statistical task in business, where it helps to inform decisions about the scheduling of production, transportation and personnel, and provides a guide to long-term strategic planning. However, business forecasting is often done poorly, and is frequently confused with planning and goals. They are three different things.

Forecasting
: is about predicting the future as accurately as possible, given all of the information available, including historical data and knowledge of any future events that might impact the forecasts.

Goals
: are what you would like to have happen. Goals should be linked to forecasts and plans, but this does not always occur. Too often, goals are set without any plan for how to achieve them, and no forecasts for whether they are realistic.

Planning
: is a response to forecasts and goals. Planning involves determining the appropriate actions that are required to make your forecasts match your goals.

Forecasting should be an integral part of the decision-making activities of management, as it can play an important role in many areas of a company. Modern organisations require short-term, medium-term and long-term forecasts, depending on the specific application.

Short-term forecasts
: are needed for the scheduling of personnel, production and transportation. As part of the scheduling process, forecasts of demand are often also required.

Medium-term forecasts
: are needed to determine future resource requirements, in order to purchase raw materials, hire personnel, or buy machinery and equipment.

Long-term forecasts
: are used in strategic planning. Such decisions must take account of market opportunities, environmental factors and internal resources.

An organisation needs to develop a forecasting system that involves several approaches to predicting uncertain events. Such forecasting systems require the development of expertise in identifying forecasting problems, applying a range of forecasting methods, selecting appropriate methods for each problem, and evaluating and refining forecasting methods over time. It is also important to have strong organisational support for the use of formal forecasting methods if they are to be used successfully.

1.3 Determining what to forecast

In the early stages of a forecasting project, decisions need to be made about what should be forecast. For example, if forecasts are required for items in a manufacturing environment, it is necessary to ask whether forecasts are needed for:

1. every product line, or for groups of products?
2. every sales outlet, or for outlets grouped by region, or only for total sales?
3. weekly data, monthly data or annual data?

It is also necessary to consider the forecasting horizon. Will forecasts be required for one month in advance, for 6 months, or for ten years? Different types of models will be necessary, depending on what forecast horizon is most important.

How frequently are forecasts required? Forecasts that need to be produced frequently are better done using an automated system than with methods that require careful manual work.

It is worth spending time talking to the people who will use the forecasts to ensure that you understand their needs, and how the forecasts are to be used, before embarking on extensive work in producing the forecasts.

Once it has been determined what forecasts are required, it is then necessary to find or collect the data on which the forecasts will be based. The data required for forecasting may already exist. These days, a lot of data are recorded, and the forecaster's task is often to identify where and how the required data are stored. The data may include sales records of a company, the historical demand for a product, or the unemployment rate for a geographic region. A large part of a forecaster's time can be spent in locating and collating the available data prior to developing suitable forecasting methods.

1.4 Forecasting data and methods

The appropriate forecasting methods depend largely on what data are available.

If there are no data available, or if the data available are not relevant to the forecasts, then **qualitative forecasting** methods must be used. These methods are not purely guesswork—there are well-developed structured approaches to obtaining good forecasts without using historical data. These methods are discussed in Chapter 6.

Quantitative forecasting can be applied when two conditions are satisfied:

1. numerical information about the past is available;

2. it is reasonable to assume that some aspects of the past patterns will continue into the future.

There is a wide range of quantitative forecasting methods, often developed within specific disciplines for specific purposes. Each method has its own properties, accuracies, and costs that must be considered when choosing a specific method.

Most quantitative prediction problems use either time series data (collected at regular intervals over time) or cross-sectional data (collected at a single point in time). In this book we are concerned with forecasting future data, and we concentrate on the time series domain.

Time series forecasting

Examples of time series data include:

- Annual Google profits
- Quarterly sales results for Amazon
- Monthly rainfall
- Weekly retail sales
- Daily IBM stock prices
- Hourly electricity demand
- 5-minute freeway traffic counts
- Time-stamped stock transaction data

Anything that is observed sequentially over time is a time series. In this book, we will only consider time series that are observed at regular intervals of time (e.g., hourly, daily, weekly, monthly, quarterly, annually). Irregularly spaced time series can also occur, but are beyond the scope of this book.

When forecasting time series data, the aim is to estimate how the sequence of observations will continue into the future. Figure 1.1 shows the quarterly Australian beer production from 2000 to the second quarter of 2010.

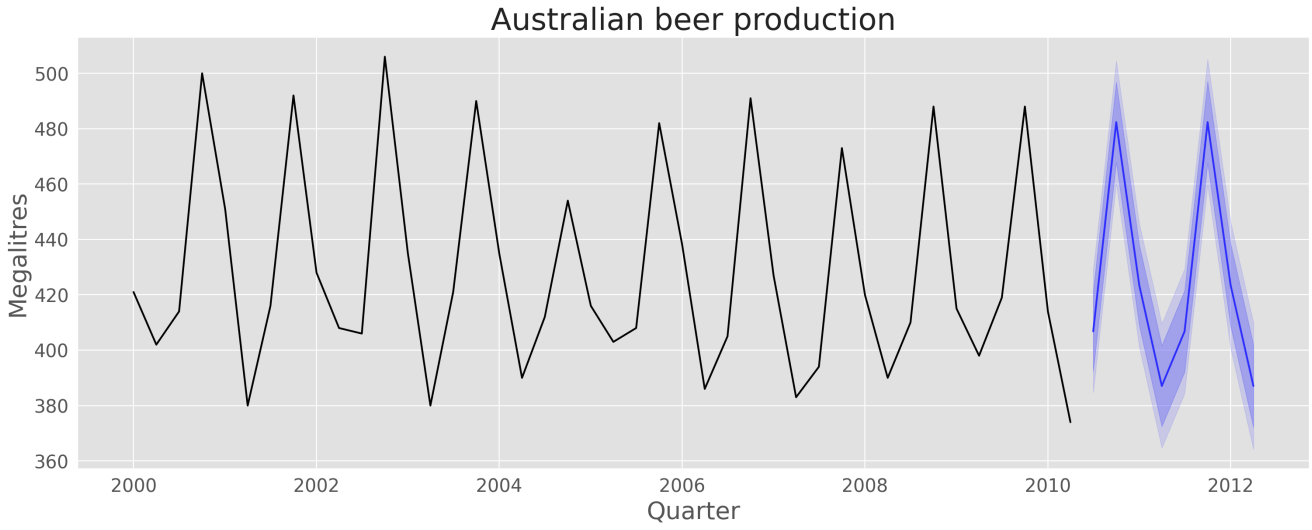

Figure 1.1: Australian quarterly beer production: 2000Q1–2010Q2, with two years of forecasts.

The blue lines show forecasts for the next two years. Notice how the forecasts have captured the seasonal pattern seen in the historical data and replicated it for the next two years. The dark shaded region shows 80% prediction intervals. That is, each future value is expected to lie in the dark shaded region with a probability of 80%. The light shaded region shows 95% prediction intervals. These prediction intervals are a useful way of displaying the uncertainty in forecasts. In this case the forecasts are expected to be accurate, and hence the prediction intervals are quite narrow.

The simplest time series forecasting methods use only information on the variable to be forecast, and make no attempt to discover the factors that affect its behaviour. Therefore they will extrapolate trend and seasonal patterns, but they ignore all other information such as marketing initiatives, competitor activity, changes in economic conditions, and so on.

Decomposition methods are helpful for studying the trend and seasonal patterns in a time series; these are discussed in Chapter 3. Popular time series models used for forecasting include exponential smoothing models and ARIMA models, discussed in Chapters 8 and 9 respectively.

Predictor variables and time series forecasting

Predictor variables are often useful in time series forecasting. For example, suppose we wish to forecast the hourly electricity demand (ED) of a hot region during the summer period. A model with predictor variables might be of the form

$$\begin{aligned}\text{ED} =& f(\text{current temperature, strength of economy, population,}\\ &\text{time of day, day of week, error}).\end{aligned}$$

The relationship is not exact — there will always be changes in electricity demand that cannot be accounted for by the predictor variables. The “error” term on the right allows for random variation and the effects of relevant variables that are not included in the model. We call this an **explanatory model** because it helps explain what causes the variation in electricity demand.

Because the electricity demand data form a time series, we could also use a time series model for forecasting. In this case, a suitable time series forecasting equation is of the form

$$\text{ED}_{t+1} = f(\text{ED}_t, \text{ED}_{t-1}, \text{ED}_{t-2}, \text{ED}_{t-3}, \dots, \text{error}),$$

where t is the present hour, $t+1$ is the next hour, $t-1$ is the previous hour, $t-2$ is two hours ago, and so on. Here, prediction of the future is based on past values of a variable, but not on external variables that may affect the system. Again, the “error” term on the right allows for random variation and the effects of relevant variables that are not included in the model.

There is also a third type of model which combines the features of the above two models. For example, it might be given by

$$\text{ED}_{t+1} = f(\text{ED}_t, \text{current temperature}, \text{time of day}, \text{day of week}, \text{error}).$$

These types of "mixed models" have been given various names in different disciplines. They are known as dynamic regression models, panel data models, longitudinal models, transfer function models, and linear system models (assuming that f is linear).

An explanatory model is useful because it incorporates information about other variables, rather than only historical values of the variable to be forecast. However, there are several reasons a forecaster might select a time series model rather than an explanatory or mixed model. First, the system may not be understood, and even if it was understood it may be extremely difficult to measure the relationships that are assumed to govern its behaviour. Second, it is necessary to know or forecast the future values of the various predictors in order to be able to forecast the variable of interest, and this may be too difficult. Third, the main concern may be only to predict what will happen, not to know why it happens. Finally, the time series model may give more accurate forecasts than an explanatory or mixed model.

The model to be used in forecasting depends on the resources and data available, the accuracy of the competing models, and the way in which the forecasting model is to be used.

1.5 Some case studies

The following four cases are from our consulting practice and demonstrate different types of forecasting situations and the associated challenges that often arise.

Case 1

The client was a large company manufacturing disposable tableware such as napkins and paper plates. They needed forecasts of each of hundreds of items every month. The time series data showed a range of patterns, some with trends, some seasonal, and some with neither. At the time, they were using their own software, written in-house, but it often produced forecasts that did not seem sensible. The methods that were being used were the following:

1. average of the last 12 months data;
2. average of the last 6 months data;
3. prediction from a straight line regression over the last 12 months;
4. prediction from a straight line regression over the last 6 months;
5. prediction obtained by a straight line through the last observation with slope equal to the average slope of the lines connecting last year's and this year's values;
6. prediction obtained by a straight line through the last observation with slope equal to the average slope of the lines connecting last year's and this year's values, where the average is taken only over the last 6 months.

They required us to tell them what was going wrong and to modify the software to provide more accurate forecasts. The software was written in COBOL, making it difficult to do any sophisticated numerical computation.

Case 2

In this case, the client was the Australian federal government, which needed to forecast the annual budget for the Pharmaceutical Benefit Scheme (PBS). The PBS provides a subsidy for many pharmaceutical products sold in Australia, and the expenditure depends on what people purchase during the year. The total expenditure was around A$7 billion in 2009, and had been underestimated by nearly $1 billion in each of the two years before we were asked to assist in developing a more accurate forecasting approach.

In order to forecast the total expenditure, it is necessary to forecast the sales volumes of hundreds of groups of pharmaceutical products using monthly data. Almost all of the groups have trends and seasonal patterns. The sales volumes for many groups have sudden jumps up or down due to changes in what drugs are subsidised. The expenditures for many groups also have sudden changes due to cheaper competitor drugs becoming available.

Thus we needed to find a forecasting method that allowed for trend and seasonality if they were present, and at the same time was robust to sudden changes in the underlying patterns. It also needed to be able to be applied automatically to a large number of time series.

Case 3

A large car fleet company asked us to help them forecast vehicle resale values. They purchase new vehicles, lease them out for three years, and then sell them. Better forecasts of vehicle sales values would mean better control of profits; understanding what affects resale values may allow leasing and sales policies to be developed in order to maximise profits.

At the time, the resale values were being forecast by a group of specialists. Unfortunately, they saw any statistical model as a threat to their jobs, and were uncooperative in providing information. Nevertheless, the company provided a large amount of data on previous vehicles and their eventual resale values.

Case 4

In this project, we needed to develop a model for forecasting weekly air passenger traffic on major domestic routes for one of Australia's leading airlines. The company required forecasts of passenger numbers for each major domestic route and for each class of passenger (economy class, business class and first class). The company provided weekly traffic data from the previous six years.

Air passenger numbers are affected by school holidays, major sporting events, advertising campaigns, competition behaviour, etc. School holidays often do not coincide in different

Australian cities, and sporting events sometimes move from one city to another. During the period of the historical data, there was a major pilots' strike during which there was no traffic for several months. A new cut-price airline also launched and folded. Towards the end of the historical data, the airline had trialled a redistribution of some economy class seats to business class, and some business class seats to first class. After several months, however, the seat classifications reverted to the original distribution.

1.6 The basic steps in a forecasting task

A forecasting task usually involves five basic steps.

Step 1: Problem definition.

Often this is the most difficult part of forecasting. Defining the problem carefully requires an understanding of the way the forecasts will be used, who requires the forecasts, and how the forecasting function fits within the organisation requiring the forecasts. A forecaster needs to spend time talking to everyone who will be involved in collecting data, maintaining databases, and using the forecasts for future planning.

Step 2: Gathering information.

There are always at least two kinds of information required: (a) statistical data, and (b) the accumulated expertise of the people who collect the data and use the forecasts. Often, it will be difficult to obtain enough historical data to be able to fit a good statistical model. In that case, the judgmental forecasting methods of Chapter 6 can be used. Occasionally, old data will be less useful due to structural changes in the system being forecast; then we may choose to use only the most recent data. However, remember that good statistical models will handle evolutionary changes in the system; don't throw away good data unnecessarily.

Step 3: Preliminary (exploratory) analysis.

Always start by graphing the data. Are there consistent patterns? Is there a significant trend? Is seasonality important? Is there evidence of the presence of business cycles? Are there any outliers in the data that need to be explained by those with expert knowledge? How strong are the relationships among the variables available for analysis? Various tools have been developed to help with this analysis. These are discussed in Chapters 2 and 3.

Step 4: Choosing and fitting models.

The best model to use depends on the availability of historical data, the strength of relationships between the forecast variable and any explanatory variables, and the way in which the forecasts are to be used. It is common to compare two or three potential models. Each model is itself an artificial construct that is based on a set of assumptions (explicit and implicit) and usually involves one or more parameters which must be estimated using

the known historical data. We will discuss regression models (Chapter 7), exponential smoothing methods (Chapter 8), Box-Jenkins ARIMA models (Chapter 9), Dynamic regression models , Hierarchical forecasting (Chapter 11), and several advanced methods including neural networks and vector autoregression (Chapter 12).

Step 5: Using and evaluating a forecasting model.

Once a model has been selected and its parameters estimated, the model is used to make forecasts. The performance of the model can only be properly evaluated after the data for the forecast period have become available. A number of methods have been developed to help in assessing the accuracy of forecasts. There are also organisational issues in using and acting on the forecasts. A brief discussion of some of these issues is given in Chapter 5. When using a forecasting model in practice, numerous practical issues arise such as how to handle missing values and outliers, or how to deal with short time series. These are discussed in Chapter 13.

1.7 The statistical forecasting perspective

The thing we are trying to forecast is unknown (or we would not be forecasting it), and so we can think of it as a *random variable*. For example, the total sales for next month could take a range of possible values, and until we add up the actual sales at the end of the month, we don't know what the value will be. So until we know the sales for next month, it is a random quantity.

Because next month is relatively close, we usually have a good idea what the likely sales values could be. On the other hand, if we are forecasting the sales for the same month next year, the possible values it could take are much more variable. In most forecasting situations, the variation associated with the thing we are forecasting will shrink as the event approaches. In other words, the further ahead we forecast, the more uncertain we are.

We can imagine many possible futures, each yielding a different value for the thing we wish to forecast. Plotted in black in Figure 1.2 are the total international arrivals to Australia from 1980 to 2015. Also shown are ten possible futures from 2016–2025.

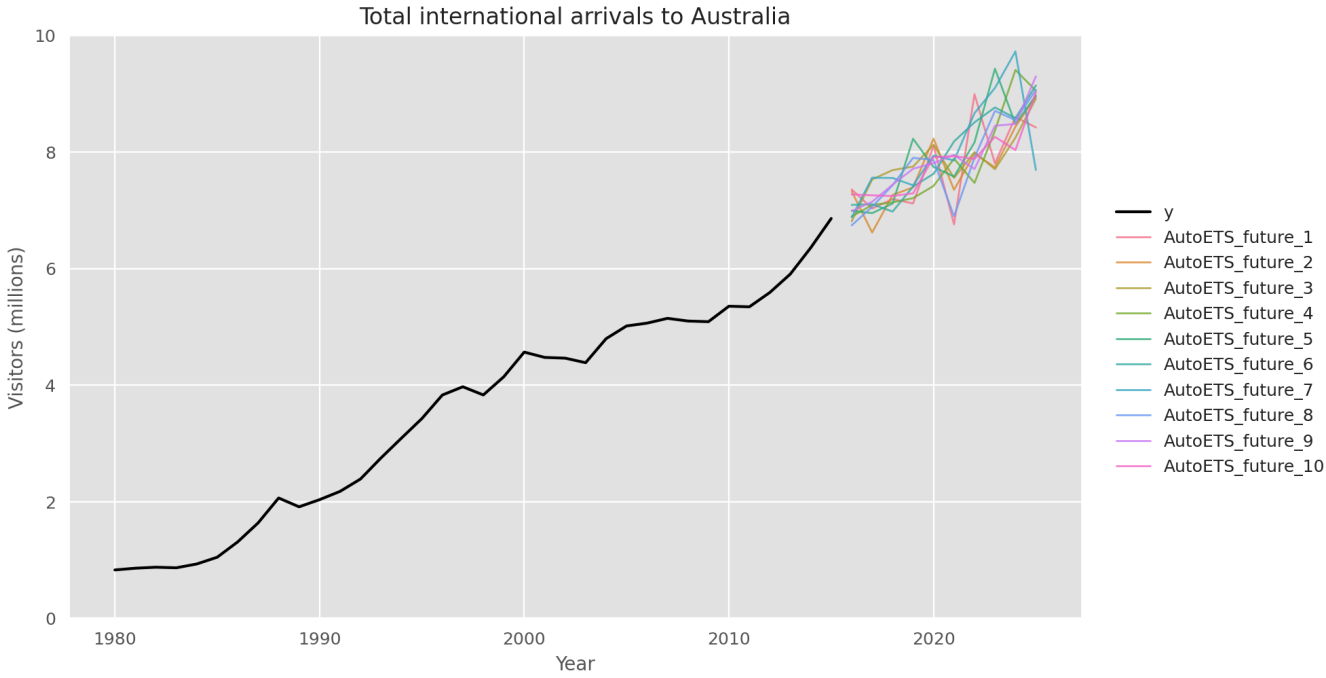

Figure 1.2: Total international visitors to Australia (1980-2015) along with ten possible futures.

When we obtain a forecast, we are estimating the *middle* of the range of possible values the random variable could take. Often, a forecast is accompanied by a **prediction interval** giving a *range* of values the random variable could take with relatively high probability. For example, a 95% prediction interval contains a range of values which should include the actual future value with probability 95%.

Rather than plotting individual possible futures as shown in Figure 1.2, we usually show these prediction intervals instead. Figure 1.3 shows 80% and 95% intervals for the future Australian international visitors. The blue line is the average of the possible future values, which we call the **point forecasts**.

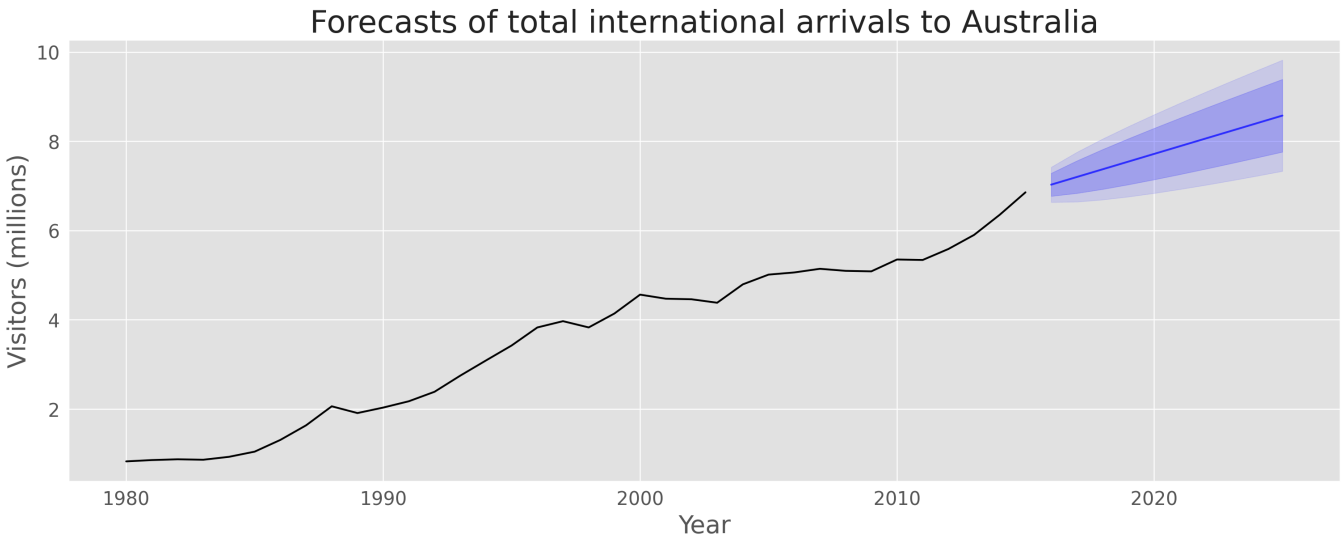

Figure 1.3: Total international visitors to Australia (1980–2015) along with 10-year forecasts and 80% and 95% prediction intervals.

We will use the subscript t for time. For example, y_t will denote the observation at time t. Suppose we denote all the information we have observed as $\mathcal{I}$ and we want to forecast y_t. We then write $y_t|\mathcal{I}$ meaning "the random variable y_t given what we know in $\mathcal{I}$". The set of values that this random variable could take, along with their relative probabilities, is known as the "probability distribution" of $y_t|\mathcal{I}$. In forecasting, we call this the **forecast distribution**.

When we talk about the "forecast", we usually mean the average value of the forecast distribution, and we put a "hat" over y to show this. Thus, we write the forecast of y_t as $\hat{y}_t$, meaning the average of the possible values that y_t could take given everything we know.

It is often useful to specify exactly what information we have used in calculating the forecast. Then we will write, for example, $\hat{y}_{t|t-1}$ to mean the forecast of y_t taking account of all previous observations $(y_1, \dots, y_{t-1})$. Similarly, $\hat{y}_{T+h|T}$ means the forecast of y_{T+h} taking account of $y_1, \dots, y_T$ (i.e., an h-step forecast taking account of all observations up to time T).

1.8 Exercises

1. For cases 3 and 4 in Section 1.5, list the possible predictor variables that might be useful, assuming that the relevant data are available.

2. For case 3 in Section 1.5, describe the five steps of forecasting in the context of this project.

1.9 Further reading

- Armstrong (2001) covers the whole field of forecasting, with each chapter written by different experts. It is highly opinionated at times (and we don't agree with everything in it), but it is full of excellent general advice on tackling forecasting problems.
- Ord, Fildes, and Kourentzes (2017) is a forecasting textbook covering some of the same areas as this book, but with a different emphasis and not focused around any particular software environment. It is written by three highly respected forecasters, with many decades of experience between them.

1.10 Used modules and classes

StatsForecast

- `StatsForecast` class

    - Core forecasting engine
- `AutoETS()` model - For automatic exponential smoothing

UtilsForecast

- `plot_series()` utility - For creating time series visualizations

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1. Codex Theodosianus 9.16.4 ↩

2. Vagrancy Act, 1824, Section 4, repealed in 1989. ↩

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